

No significant difference in clinical outcome and knee stability between patellar tendon and semitendinosus tendon in anterior cruciate ligament reconstruction

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Abstract

Introduction ACL reconstruction with either patellar tendon or semitendinosus tendon autografts are standard procedures. Between these two grafts might be differences in stability, morbidity, or long-term changes. This study investigates outcomes of ACL reconstruction with patellar tendon versus semitendinosus tendon autografts. We hypothesize no significant differences in clinical outcome and knee stability between both groups.

Methods In a randomized prospective trial, we operated 62 ACL-deficient patients, 45 males and 17 females with a mean age of 29.8 years (min. 18, max. 44). We reconstructed the ligament using either autologous patellar tendon ($n = 31$) or semitendinosus tendon ($n = 31$). After 10 years of follow-up, we investigated 47 patients of the study. For evaluation we used a standard clinical examination including one-leg jump test and KT-1000 instrumental translation measure, visual analog pain scale, IKDC subjective knee form, Lysholm score, Tegner activity scale, and standard X-rays of the knee.

Results The data did not show any significant differences between the two groups. Between 5 and 10 years after ACL reconstruction both groups started to develop degenerative arthritic changes, which were detectable in standard radiographs of the knee. At 10-year follow-up mean IKDC for the BPTB group was 1.8 (min. 1, max. 3) and for the ST group it was 2.2 (min 1, max. 4), $p = 0.35$. Regarding Tegner activity scale after 10 years, the BPTB

group showed a mean score of 5.9 (min. 4, max. 9) versus 5.1 (min. 3, max. 7) in the ST group, $p = 0.53$. For the Lysholm score the BPTB group reached a mean of 92.0 (min. 63, max. 98) and the ST group 91.8 (min. 62, max. 98) points, $p = 0.66$. There is a tendency for higher donor site morbidity in the BPTB group than in the ST group, $p = 0.07$.

Conclusions Both, patellar tendon and semitendinosus tendon are safe autografts for ACL reconstruction. Regarding graft selection, individual patient-dependent factors should be considered. ACL reconstruction cannot fully restore pre-injury status of knee joint function in the majority of cases.

Keywords Anterior cruciate ligament (ACL) · Patellar tendon · Semitendinosus tendon · Knee · Instability · Arthritis

Introduction

Anterior cruciate ligament (ACL) tears are common injuries in sports, especially in soccer players and skiers. Typical trauma mechanism is flexion, external rotation, and valgus stress at the same time. Often there are additional injuries to the medial collateral ligament and meniscus. Unstable knees are prone to secondary pathologies like medial or lateral meniscal tears and arthritis [1].

In young and active patients, ACL reconstruction is common treatment. For reconstruction, autologous semitendinosus or patellar tendon is used. Earlier studies have shown those grafts to be safe with good clinical results and without major differences in outcome [2, 3]. There is a tendency for elevated pain at donor site after harvesting the mid-third of the patellar tendon in bone–tendon–bone

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technique, especially when the patient gets on his knees [4]. Potential long-term differences in outcomes after ACL reconstruction via autologous patellar tendon versus semitendinosus tendon are not definitely known since there are few randomized prospective long-time follow-up studies available.

We conducted a randomized prospective trial with 10 years of follow-up to detect differences in outcomes after ACL reconstruction using either autologous patellar or semitendinosus tendon graft. We examined both groups 1, 5, and 10 years postoperatively to also detect possible changes of graft laxity over time. We hypothesize no significant differences in clinical outcome and knee stability between the two groups.

Patients and methods

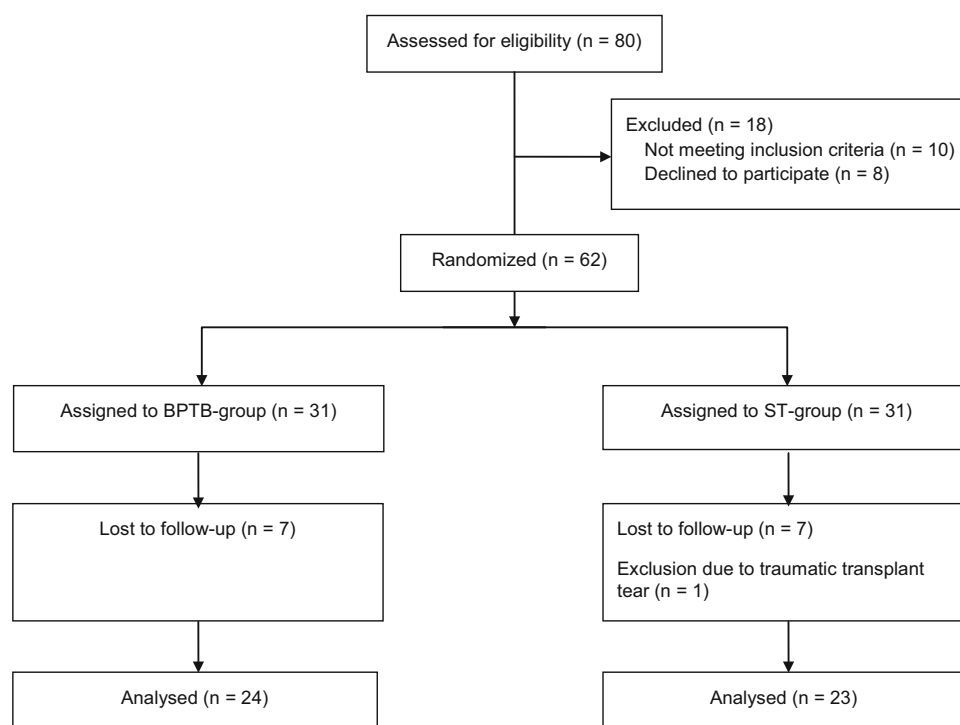
BPTB bone–patellar tendon–bone, *ST* semitendinosus tendon

In our institution between November 1998 and November 1999, 80 consecutive patients received ACL reconstruction by the senior author. In a prospective randomized trial preoperatively we included traumatic ACL tears with clinical instability. Exclusion criteria were bilateral injuries, knee luxation, fracture, tear of the posterior cruciate ligament or posterolateral instability, and operations of the same joint in the past. Patients were randomized into two

groups via drawing of an envelope (Fig. 1). All patients received ACL reconstruction using either the middle third of the ipsilateral patellar tendon in bone–patellar tendon–bone technique (BPTB group) with interference screw fixation at femoral and tibial or ipsilateral semitendinosus tendon (ST group), with endobutton fixation at the femur and suture-disc fixation at the tibia. The semitendinosus tendon was prepared as a quadruple or triple graft according to its length. Most ACL tears happened due to trauma during sports activities like soccer (53.2 %) or ski (30.0 %). There were ten isolated ACL tears. The other patients had additional knee joint lesions. Those were meniscal ($n = 40$), chondral ($n = 38$), and tears of the medial collateral ligament ($n = 8$). Accompanying injuries were treated simultaneously with ACL reconstruction. Out of 62 patients operated in the trial, 45 (72.6 %) were males and 17 (27.4 %) females. The ratio of males/females in the BPTB group was 22/9 and in the ST group 23/8. The mean age of all patients was 29.8 (min. 18, max. 44) years. A covariant analysis showed that there is no significant difference between groups regarding patients' age, sex, accompanying lesions, or time between injury and operation.

We used the same postoperative rehabilitation protocol in both groups. For 6 weeks patients were allowed full weight bearing when standing and partial weight bearing with 20 kg during walking with a knee brace. Range of motion was limited to Ext/Flex 0/0/90°. Isometric training of the active dynamic knee stabilizing muscles was

Fig. 1 CONSORT flow diagram for the present trial



performed. After a clinical control in our outpatient clinic 6 weeks postoperatively, patients were allowed full weight bearing without knee brace and full mobilization of the knee. Return to competitive sports was allowed after 9 months, if muscle strength and neuromuscular control was reestablished.

At follow-up evaluations in our outpatient clinic 1, 5, and 10 years postoperatively we did a standard clinical investigation including Lachman test, pivot-shift test, range of motion, thigh circumference measure (measured 10 cm proximal of medial knee joint line), one-leg jump test [5], and instrumental translation measure with the KT-1000 arthrometer (MEDmetric, San Diego, CA, USA) [6]. Additionally, we used the visual analog pain scale (VAS) [7], IKDC score [8–12], Lysholm score [13], Tegner activity scale [14], and standard X-rays of the knee. For radiological evaluation during follow-up, we used the score of Sherman [15, 16].

For statistical analysis, we used the SPSS 19.0 program (SPSS Inc., USA). A p value of <0.05 was considered to be statistically significant. After testing for normal distribution, we used the t test for normally distributed data or the non-parametric Mann–Whitney U test for not normally distributed data.

Results

Ten years after ACL reconstruction, we lost 14 patients to follow-up; five could not be located and nine refused to attend the follow-up because of business reasons or unwillingness. We excluded one patient (ST group) from the study due to traumatic transplant tear 3 years after index ACL reconstruction. For revision surgery, autologous BPTB transplant was used. The dropout rate was 24.2 % (Fig. 1).

There was one early extraarticular wound infection in the ST group. It was treated successfully by revision surgery after 5 days with open debridement, irrigation, and antibiotic treatment for 14 days.

At 10 years of follow-up, patients in the BPTB group had higher donor site morbidity than the ST group ($p = 0.07$). In the BPTB group, most patients had no (34.4 %) or low (43.6 %) pain, 13.2 % had moderate and 8.8 % had severe pain. In the ST group, most patients had no (57.3 %) or low (38.3 %) pain, 4.4 % had moderate and not any patient had severe pain. One patient in the BPTB group had an extension deficit of 5° . All other patients had normal range of motion.

At 10 years of follow-up, 22 (46,8 %) patients had a reduction of thigh circumference (measured 10 cm proximal of medial knee joint line) of more than 1 cm in comparison with the contralateral thigh. Eight of those patients

were in the BPTB group and 14 in the ST group ($p = 0.13$).

The Lachman test did not show significant differences between the two groups (Table 1).

Table 2 shows the difference between the patient's uninjured knee and the operated knee regarding anterior translation of the tibia measured with the KT-1000 arthrometer at different forces 10 years after ACL reconstruction. The difference between BPTB group and ST group was not significant.

The mean IKDC score (Table 3) 1 year postoperatively in the BPTB group was 2.2 (min. 1, max. 3); in the ST group it was 2.6 (min. 1, max. 4); after 5 years it was 1.7 (min. 1, max. 3) and 1.7 (min. 1, max. 4), respectively. The mean IKDC score after 10 years in the BPTB group was 1.8 (min. 1, max. 3); in the ST group it was 2.2 (min. 1, max. 4). The difference was not significant ($p = 0.35$). The mean Tegner activity scale after 10 years in the BPTB group was 5.9 (min. 4, max. 9), in the ST group 5.1 (min. 3, max. 7). The difference was not significant ($p = 0.53$). Regarding the Lysholm score, the BPTB group reached a mean of 92.0 (min. 63, max. 98) and the ST group 91.8 (min. 62, max. 98) points ($p = 0.66$). Two patients of each group complained about slight giving-way sometimes. All other patients did not feel giving-way symptoms in sports or activities of daily living at 10 years of follow-up.

In the single-leg jump test, the BPTB group reached 9 cm less distance with the operated leg ($149 \text{ cm} \pm 7.3$) than with the uninjured leg ($158 \text{ cm} \pm 6.5$). In the ST group this difference was 13 cm ($137 \text{ cm} \pm 7.1$ with the operated and $150 \text{ cm} \pm 6.9$ with the uninjured leg). There is not a significant difference in comparison of both groups ($p = 0.33$).

Table 3 shows radiological and clinical outcome parameters preoperatively and at time of follow-up visits 1, 5, and 10 years after ACL reconstruction.

Ten years after ACL reconstruction patients in the BPTB group were very satisfied (73.1 %), satisfied (26.9 %), or unsatisfied (0 %), whereas patients in the ST group were very satisfied (80.1 %), satisfied (11.5 %), or unsatisfied (7.8 %).

We did not find any significant differences between groups during a 10-year follow-up after ACL reconstruction using either BPTB or ST autologous grafts.

Table 1 Lachman test ten years after ACL reconstruction in the BPTB group and the ST group ($p = 0.19$)

Lachman test (mm)	BPTB group (%)	ST group (%)
0–2 (negative)	16 (66.7)	12 (52.2)
3–5 (+)	8 (33.3)	9 (39.1)
6–10 (++)	0 (0)	1 (4.4)
>10 (+++)	0 (0)	1 (4.4)

BPTB bone–patellar tendon–bone, ST semitendinosus tendon

Table 2 Anterior translation of the tibia in relation to the uninjured side, measured with the KT-1000 arthrometer at 10-year follow-up of BPTB group and ST group ($p = 0.51$)

Δ KT-1000 (mm)	BPTB group (%)	ST group (%)
89 N		
<3	21 (87.5)	19 (82.6)
3–6	3 (12.5)	4 (17.4)
6.5–10	0 (0)	0 (0)
>10	0 (0)	0 (0)
101 N		
<3	19 (79.2)	18 (78.3)
3–6	5 (20.8)	4 (17.4)
6.5–10	0 (0)	1 (4.4)
>10	0 (0)	0 (0)
Max. force		
<3	19 (79.2)	18 (78.3)
3–6	5 (20.8)	5 (21.7)
6.5–10	0 (0)	0 (0)
>10	0 (0)	0 (0)

BPTB bone–patellar tendon–bone, ST semitendinosus tendon

Table 3 Radiological and clinical outcome indicators over a period of 10 years after ACL reconstruction

Score	Pre-surgery	1 year	5 years	10 years
Sherman <i>T</i>				
BPTB	25.9	25.9	25.5	22.2
ST	25.2	25.2	24.9	21.5
IKDC				
BPTB	3.9	2.2	1.7	1.8
ST	3.7	2.6	1.7	2.2
Lysholm				
BPTB	55.4	89.2	88.3	92.0
ST	65.2	89.3	88.8	91.8
Tegner				
BPTB	7.4	6.3	5.6	5.9
ST	6.8	5.3	5.3	5.1

BPTB bone–patellar tendon–bone, ST semitendinosus tendon

Discussion

We found no significant differences in clinical outcome and knee stability between patellar tendon and semitendinosus tendon autografts in anterior cruciate ligament reconstruction.

There are only few randomized prospective long-time follow-up studies available comparing results after ACL reconstruction with different grafts. In this randomized prospective study we analyzed the outcome of ACL reconstruction using either autologous patellar tendon or

semitendinosus tendon graft over a follow-up period of 10 years. Overall, the outcome of our patients in both groups is similar to reported outcome in the literature [2, 17]. Despite insignificant differences between both groups, there is a strong tendency for higher donor site morbidity in the BPTB group in comparison with the ST group ($p = 0.07$). Those findings have also been reported by other authors [2, 18].

During the first 5 years after surgery, patients in both groups improved in the IKDC score (Table 3). The ST group needed more time to reach the best score after 5 years. At 5 years of follow-up, mean IKDC scores of both groups were identical (1.7). Comparing IKDC scores 5 and 10 years post-surgery, we can see a tendency of worsening over time, especially in the ST group, but this is not significant. It could be explained by detectable radiological degenerative changes after 10 years of follow-up. While the Sherman *T* score did not change from time of surgery until 5-year follow-up examination, there is a tendency of decreasing *T* score from 5 to 10 years post-surgery. This is similar in both groups and indicates beginning of posttraumatic arthritis. These findings are congruent with a systematic review from the year 2015 [19]. Here, Spahn et al. conclude detectable degenerative changes in radiographs 5 years after ACL reconstruction. According to the review, early arthritic changes are asymptomatic after 5 years of follow-up. As posttraumatic arthritis progresses, it usually becomes symptomatic from the eighth year onwards. Struwer et al. found that after ACL reconstruction with semitendinosus tendon, 25 % of patients developed symptomatic higher degree osteoarthritis. These relevant degenerative changes significantly correlate with long-term knee joint instability [20].

In the literature as in our study ACL reconstruction could not establish the same stability of the uninjured contralateral side in most cases regardless of graft choice. This seems to be a problem especially for patients with high demands. There are two patients with bad results with a Lysholm score under 65 in each group. This could be explained by persistent clinically relevant anterior instability in these patients.

Potential shortcomings of the study include the limited number of patients and the relatively high dropout rate. Analyzing the degenerative changes of the knees after ACL reconstruction versus a conservatively treated control group would be interesting for future research.

Conclusions

Both, patellar tendon and semitendinosus tendon are safe autografts for ACL reconstruction. We found no significant differences between the BPTB group and the ST group.

There was a tendency for more persistent pain at donor site in the BPTB group. Individual patient-related factors like kneeling should be considered in graft selection.

Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

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